Kely Tamara Romero Romero

Junior Data Analyst | Python, Machine Learning, AI Automation | Agronomist

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PROFESSIONAL SUMMARY

Junior data analyst and Agronomist (UNL, 2026). Built regression and machine learning models in Python (pandas, scikit-learn), R and InfoStat to estimate coffee yield and quality, reaching R^2 of 0.87 and 0.98. 2026 training in AI for developers: Claude API, RAG, AI agents, MCP, OpenAI Codex and workflow automation with n8n and Python. Seeking a role in data analysis, automation or applied AI, preferably remote.

SKILLS

Data analysis: Python (pandas, NumPy, scikit-learn, matplotlib, seaborn), R (tidyverse, ggplot2, sf), InfoStat, Excel, Jupyter; linear, multiple and polynomial regression, Random Forest, cross-validation, EDA, data cleaning, inferential statistics

AI, automation and development: Claude Code, Claude API, RAG, AI agents, agentic design patterns, MCP, OpenAI Codex, Spec-Driven Development, n8n, workflow automation with Python, prompt engineering, Node.js (basic), Git, GitHub, Visual Studio Code

Agricultural domain: coffee and cocoa cultivation and post-harvest, soil and cup quality analysis, work with farmers

Languages: Spanish (native) | English intermediate B1 (CEFR) – UNL Language Institute, 768 h

EXPERIENCE

Thesis Researcher – Data Analysis and Machine Learning | 2024 – 2026
Universidad Nacional de Loja | Loja, Ecuador

Project: Statistical models to estimate coffee yield and cup quality in the province of Loja.

- Built multiple linear regressions in InfoStat for yield (R^2 = 0.87) and cup quality (R^2 = 0.98) with data from 11 farms.
- Programmed Ridge polynomial regression with cross-validation and Random Forest (LOOCV) in Python (pandas, NumPy, scikit-learn).
- Performed data cleaning, exploratory data analysis (EDA), correlation matrices and partial residual plots (matplotlib, seaborn); created study area maps in R (ggplot2, sf).
- Identified fertilization, planting density and altitude as the strongest yield drivers, and fruit pH and soil electrical conductivity as key cup quality factors.
- Designed a combined agro-productive and sensory performance index to compare farms; the best farm reached 17.8 qq/ha and 86.38 points (SCA protocol).
- Gathered information from farmers, collected coffee and soil samples, analyzed pH, titratable acidity and soluble solids in the lab, and coordinated cupping with Q Arabica graders (funded by UNL).

Intern – Coffee and Cocoa Production and Post-harvest | Mar 2024 – Apr 2024 | 360 h pre-professional internship
APEOSAE – Federation of Organic Exporters of the Southern Ecuadorian Amazon | Nangaritza, Zamora Chinchipe

- Managed a Geisha coffee nursery and performed side-veneer grafting on cocoa (30 seedlings in charge).
- Supported post-harvest: cocoa drying; sorting, roasting, grinding and packaging of organic coffee.

Research Assistant | Apr 2022 – Oct 2022
Universidad Nacional de Loja | Loja, Ecuador

- Morphological identification project of native cherimoya populations with agronomic potential: processed and analyzed data and lab samples for their morphological characterization.

EDUCATION

B.Sc. Agronomic Engineering (degree obtained) | Universidad Nacional de Loja – Faculty of Agriculture | Sep 2026

COURSES AND CERTIFICATIONS

- **AI and agents – DevTalles (2026):** AI for Developers: Claude API, RAG and Agents with Node • Agentic design patterns • Codex: agents, skills, MCP and Spec-Driven Development • Claude Code • Prompt engineering
- **Programming and automation – DevTalles (2026):** Python: Applied artificial intelligence • Python + n8n: Automate everyday routines • GIT+GitHub • Visual Studio Code
- Descriptive and Inferential Statistics Workshops – GEM Training & Consulting (2025) and Florida Global University (2026)

AWARDS

- Academic Excellence Incentive – Universidad Nacional de Loja (Oct 2024 – Mar 2025).
- Scientific poster presenter – III Formative Research Conference in Agronomy (February 2025).